MOTIVATION LETTER

I have recently completed my Bachelor's degree in Computer Science and Engineering at the National Institute of Textile Engineering and Research (NITER), a constituent institute of the University of Dhaka. I have already sat for my final, eighth-semester examinations and am currently awaiting the official release of my result and I am applying for the Government Scholarship of the Czech Republic to pursue a Follow-up Master's degree in Artificial Intelligence and Data Processing (Study Programme Code: N0612A140006) at Masaryk University. My interest in this specific programme is not general while researching Czech universities on the offered list, I found that Masaryk's Faculty of Informatics hosts a research group (RationAI) working on explainable, privacy-preserving federated machine learning for medicine, which is precisely the direction my own undergraduate research has already taken. My goal is to build on that foundation during my Master's, and to return the experience and skills gained to Bangladesh's still-developing AI and healthcare-technology sector, and to the Czech Republic through continued collaboration afterward.

My academic path has not been easy. My father passed away when I was young, and growing up as the youngest of six siblings watching my family sacrifice for my education taught me to treat every opportunity as earned, not assumed which is why I pushed toward research rather than a purely coursework-based degree. At NITER, I maintained approximately **91.5% academic performance** through my seventh semester (final result pending), while conducting independent research and taking on roles including selection for the **2026 Aspire Leaders Program**, Delegate at the **IMUN Online Conference (Thailand),** Round 2 qualifier at the **SOLVIO AI Hackathon**, hands-on training in computer-aided drug design at **ABCD Laboratory Bangladesh**, and volunteer work with **Shikkhar Alo**.

Alongside my studies, I have built a genuine research and professional track record in machine learning. I authored the undergraduate thesis *LiteGAN-FedNet: Privacy-Preserving and **Explainable Federated Learning** for Brain Tumor MRI Classification*, and, because no adequate clinical dataset existed for this work, I personally coordinated the ethically cleared collection of a new Bangladeshi brain-tumor MRI dataset, which I published openly in *Data in Brief* so that other researchers could reuse it. My research work has also been recognized internationally, with my papers accepted at the Conference on Computer Vision and Pattern Recognition **(CVPR)** and the International Conference on Artificial Intelligence and Statistics **(AISTATS)**, two leading international conferences in computer vision, machine learning, and artificial intelligence. I have also worked on **quantum-inspired** feature engineering for non-invasive glucose-level monitoring system from Raman spectroscopy. Professionally, I served for **one year as a Research Assistant at the Institutional Research & Consultancy Board (IRCB)**, currently work as a **Junior Researcher at Research Ustad**, and completed a Campus Ambassador internship with the International Model United Nations **(IMUN)** that has taught me to communicate technical work to non-technical audiences, a skill I consider essential for any researcher who wants their work actually used.

I want this Master's specifically because my undergraduate work has already shown me its own limits: I have simulated federated learning on a single machine, and I need proper training in distributed systems and large-scale data infrastructure to make it real; I have built explainability tools without formal grounding in the theory behind them; and, as a student from a developing country, I have carried out all of this work without access to a dedicated research lab, GPU cluster, or resident research group every result so far was built with borrowed compute and self-taught theory. **Masaryk's programme**, and **RationAI** in particular co-founded by **doc. RNDr. Tomáš Brázdil, Ph.D.**, whose research applies machine learning to digital pathology and who continues to supervise thesis work on explainable AI in medical imaging offers exactly the laboratory environment and mentorship my work has been missing. In return, I believe I can contribute a direction the group has not yet pursued: extending its federated and explainable methods from 2D pathology imaging to volumetric radiology, and from well-resourced clinical settings to the low-bandwidth, data-scarce conditions typical of hospitals in my own country.

I do not intend to remain abroad after this degree. Bangladesh currently has almost no home-grown research infrastructure for privacy-preserving medical AI, and I want to be part of building it. I am deeply grateful for the opportunity to be considered for this scholarship it would mean more to me, and to my family, than I can easily put into a page and I would carry that responsibility seriously through every year of study, and back to my country afterward. Thank you for your time and consideration.